# Soumya Snigdha Kundu

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## Education

**King's College London** — Ph.D. Biomedical Engineering and Imaging Science — Oct 2023 – Oct 2027
**Queen Mary University of London** — M.Sc. Machine Learning (Distinction) — Sep 2022 – Sep 2023
**SRM Institute of Science and Technology** — B.Tech. Computer Science (Distinction) — Jul 2018 – May 2022

## Representative Papers

Hsieh, I., **S. S. Kundu**, Vercauteren, T., & Dorent, R. (2027). SinkSLOT: Sinkhorn via Sparse Lifted Optimal Transport. **AAAI**
Großkopf, E.*, **S. S. Kundu*** et al. (2026). Redefining Instance Matching: A Unified Framework for Part-Aware Matching in Panoptic Segmentation Evaluation. **NeurIPS**
**S. S. Kundu** et al. (2026). Instance Awareness of Multi-class Semantic Segmentation Loss Functions. *CV4CLINIC@***CVPR**
Naidu, R. & **S. S. Kundu** (2021). Improved variants of Score-CAM via Smoothing and Integrating. *RCV@***CVPR**

## Relevant Industry Experience

**Cosine (YC 23)** — Machine Learning and Product Intern (Genie 3; Multilingual SWE-Bench) — Summer 2025

## Representative Software & Open Source Contributions

**Panoptica** — [Maintainer] Instance-wise segmentation metrics — 33⋆
**Ultraflwr** — [Author] Federated learning on the edge — 30⋆
**torchsparsegradutils** — [Maintainer] Utilities for PyTorch sparse tensors — 40⋆
**MONAI** — [Maintainer] Medical Open Network for AI — 8.8k⋆
**cellfinder** — [Maintainer] Cell detection in whole-brain microscopy (BrainGlobe) — 232⋆

## Representative Fellowships & Grants

Google Summer of Code (**GSoC**) — Neuroinformatics Unit (1,141 of 23,371 proposals selected, ∼4.9%) — 2026
Claude for Open Source — Anthropic (open source maintainer grant, $1,200) — 2026
Codex for Open Source — OpenAI (open source maintainer grant, $1,200) — 2026
MRC DTP Postgraduate Studentship — King's College London (<2% selection, ∼£205k) — 2023
BDI Summer Internship — University of Oxford (1 of 4 selected) — 2023
Summer Research Internship — IIT Gandhinagar (<0.008% selection) — 2023
**MITACS** Globalink Research Internship — TRIUMF (UVic) (<3% selection) — 2021

## Representative Awards

MICCAI'26 BraTS-METS Challenge — 3rd Place — 2026
SPIE Medical Imaging Travel Grant Award ($1000) — 2025
IEEE ISBI'24 Best Student Poster Award Finalist (Top 8 of 717) — 2024
UKRI Fast Start: Innovation Grant — Co-Applicant (∼£50k) — 2022

## Programming & Software Development

**Languages:** Python, C++, Bash, LaTeX
**ML:** PyTorch, JAX, NumPy, CUDA
**Tools:** Git, Slurm, Docker, Kubernetes, macOS, Linux

## Graduate Teaching

Image and Signal Processing (Prof. James Housden) — Spring 2026
Python (Dr. Marc Modat and Dr. Jonathan Shapey) — Spring 2026
Journal Club — Biomedical Engineering (Prof. Monica Agromayor) — Winter 2024 & Spring 2025
Statistics — Biomedical Engineering (Prof. James Housden) — Spring 2025

## Supervision

Yingfang Tao — KCL BSc/MSc Thesis Supervision — 2025–27
Jai Pradeep Awasthi and Pranav Rustagi (LNMIIT) with Yang Li — Summer 2025
In2Stem (4 High School students) with Lorena Garcia-Foncillas Macias — Summers 2024/25/26

## Invited Talks

Adapting cellfinder to single-channel and 2.5-D data (NIU Open Software Summer School, SWC London, UCL.) — 2026
Exploring Instance Imbalance in Medical Semantic Segmentation (Voxel51 Visual AI in Healthcare Meetup) — 2024

## Academic Service

**Programme Committee:** BrainWorks @ MICCAI 2026
**Reviewing:** MICCAI, ICML, NeurIPS, ICLR, AISTATS, IEEE-ISBI.

References available on request.